



CPE 342

Microprocessor Systems & Interfacing

8254 Programmable Interval Timer

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8254 timer/counter interfacing with 8088 CPU

- Outline
 - 8254 Timer/Counter
 - Interfacing with 8088

8254 timer/counter

- The Intel 8254 is a counter/timer device designed to solve the common timing control problems in microcomputer system design.
- It provides three independent 16-bit counters, each capable of handling clock inputs up to 10 MHz.
- Compatible with All Intel and Most other Microprocessors
- The 8254 is a programmable interval timer/counter designed for use with Intel microcomputer systems.
- It is a general purpose, multi-timing element that can be treated as an array of I/O ports in the system software.

8254 timer/counter

- The 8254 solves one of the most common problems in any microcomputer system, the generation of accurate time delays under software control. Instead of setting up timing loops in software, the programmer configures the 8254 to match his requirements and programs one of the counters for the desired delay.

Pinout of 8254 Timer/Counter

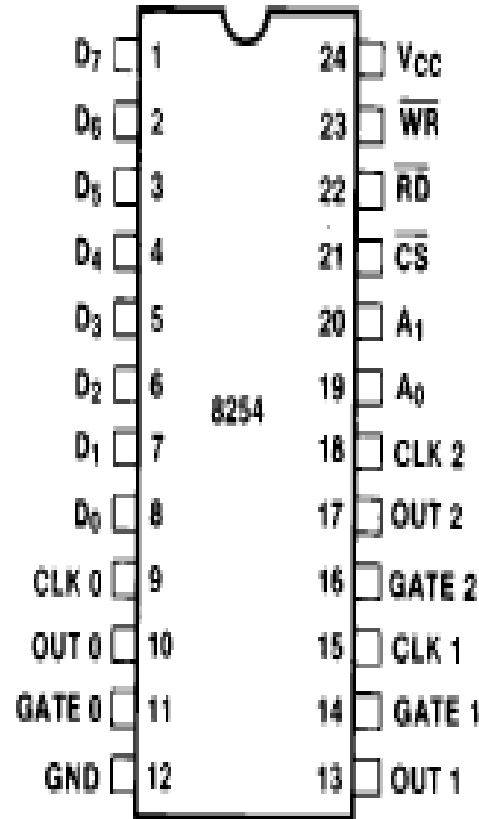
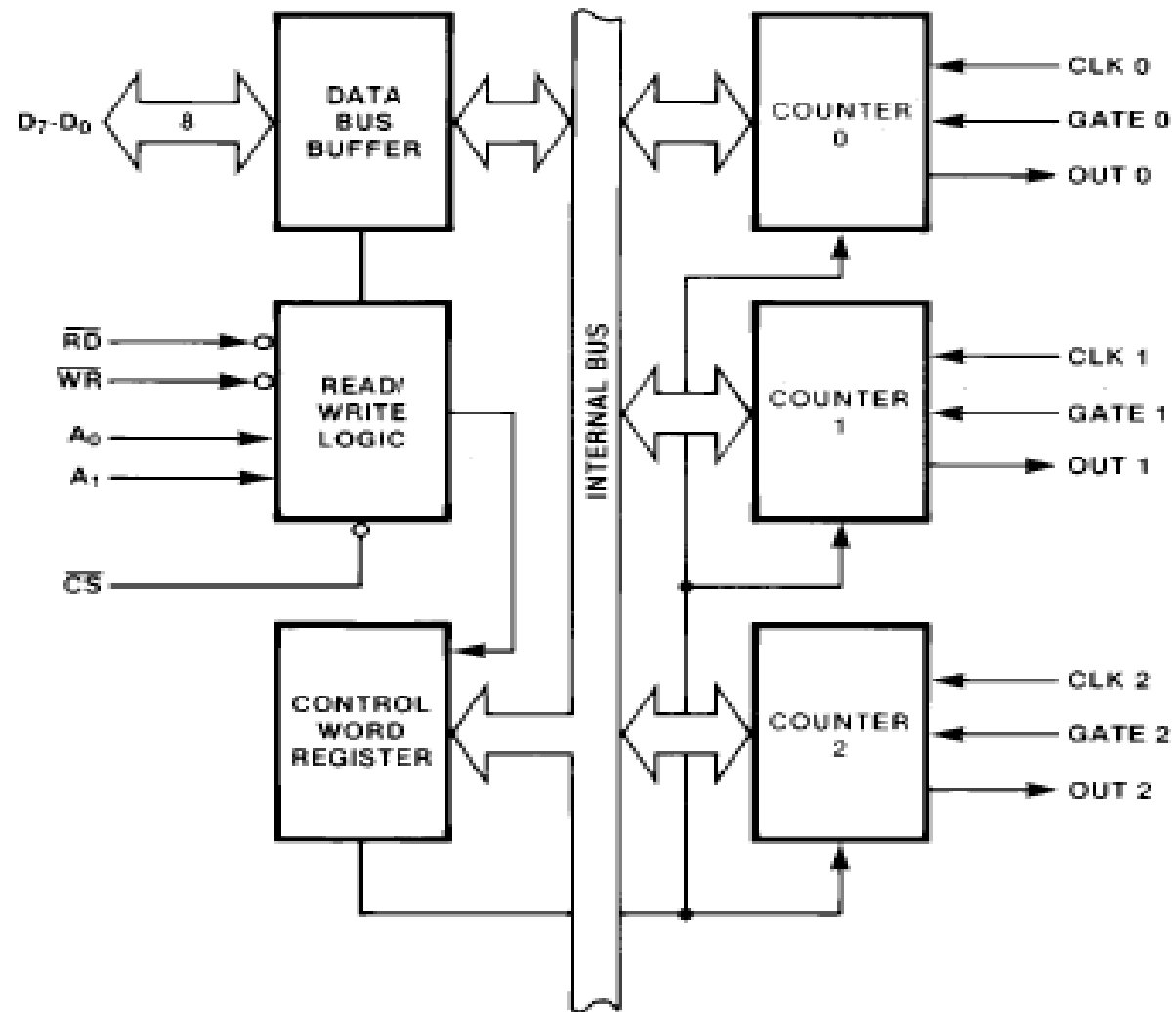


Table 1. Pin Description

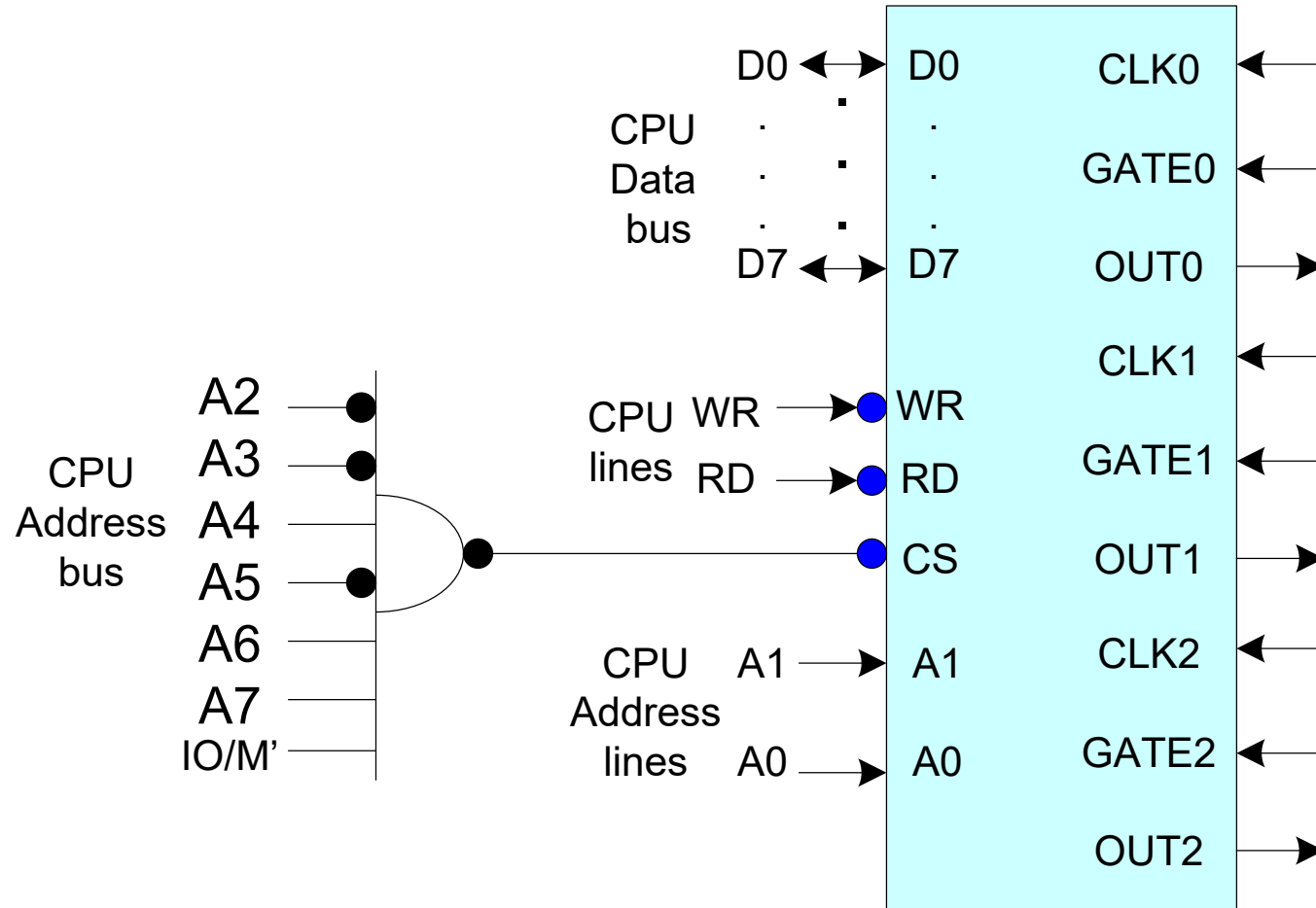
Symbol	Pin No.	Type	Name and Function		
D ₇ -D ₀	1-8	I/O	DATA: Bi-directional three state data bus lines, connected to system data bus.		
CLK 0	9	I	CLOCK 0: Clock input of Counter 0.		
OUT 0	10	O	OUTPUT 0: Output of Counter 0.		
GATE 0	11	I	GATE 0: Gate input of Counter 0.		
GND	12		GROUND: Power supply connection.		
V _{CC}	24		POWER: + 5V power supply connection.		
WR	23	I	WRITE CONTROL: This input is low during CPU write operations.		
RD	22	I	READ CONTROL: This input is low during CPU read operations.		
CS	21	I	CHIP SELECT: A low on this input enables the 8254 to respond to RD and WR signals. RD and WR are ignored otherwise.		
A ₁ , A ₀	20-19	I	ADDRESS: Used to select one of the three Counters or the Control Word Register for read or write operations. Normally connected to the system address bus.		
			A₁	A₀	Selects
			0	0	Counter 0
			0	1	Counter 1
			1	0	Counter 2
			1	1	Control Word Register
CLK 2	18	I	CLOCK 2: Clock input of Counter 2.		
OUT 2	17	O	OUT 2: Output of Counter 2.		
GATE 2	16	I	GATE 2: Gate input of Counter 2.		
CLK 1	15	I	CLOCK 1: Clock input of Counter 1.		
GATE 1	14	I	GATE 1: Gate input of Counter 1.		
OUT 1	13	O	OUT 1: Output of Counter 1.		



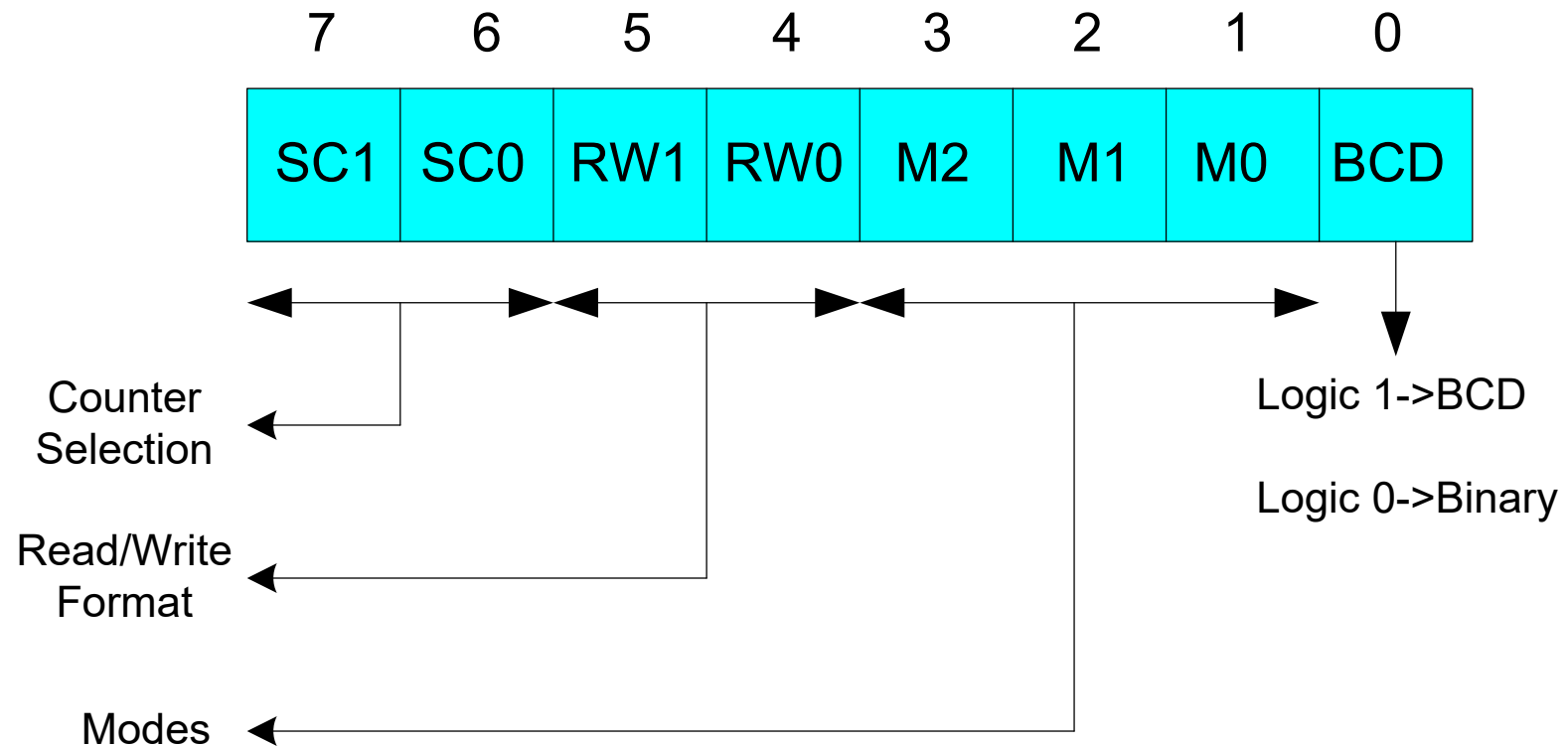
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Figure 1. 8254 Block Diagram

8254 Interfacing with 8088 CPU



Count word Register



8254



CS'	RD'	WR'	A1	A0	OPERATION
0	1	0	0	0	WRITE COUNTER 0
0	1	0	0	1	WRITE COUNTER 1
0	1	0	1	0	WRITE COUNTER 2
0	1	0	1	1	WRITE CONTROL WORD
0	0	1	0	0	READ COUNTER 0
0	0	1	0	1	READ COUNTER 1
0	0	1	1	0	READ COUNTER 2
0	0	1	1	1	NO OPERATION (TRISTATED)
0	1	1	X	X	NO OPERATION (TRISTATED)
1	X	X	X	X	8254 NOT SELECTED

Count word Register

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
SC1	SC0	RW1	RW0	M2	M1	M0	BCD

SC—Select Counter
SC1 SC0

0	0	Select Counter 0
0	1	Select Counter 1
1	0	Select Counter 2
1	1	Read-Back Command (see Read Operations)

RW—Read/Write
RW1 RW0

0	0	Counter Latch Command (see Read Operations)
0	1	Read/Write least significant byte only
1	0	Read/Write most significant byte only
1	1	Read/Write least significant byte first, then most significant byte

M—Mode
M2 M1 M0

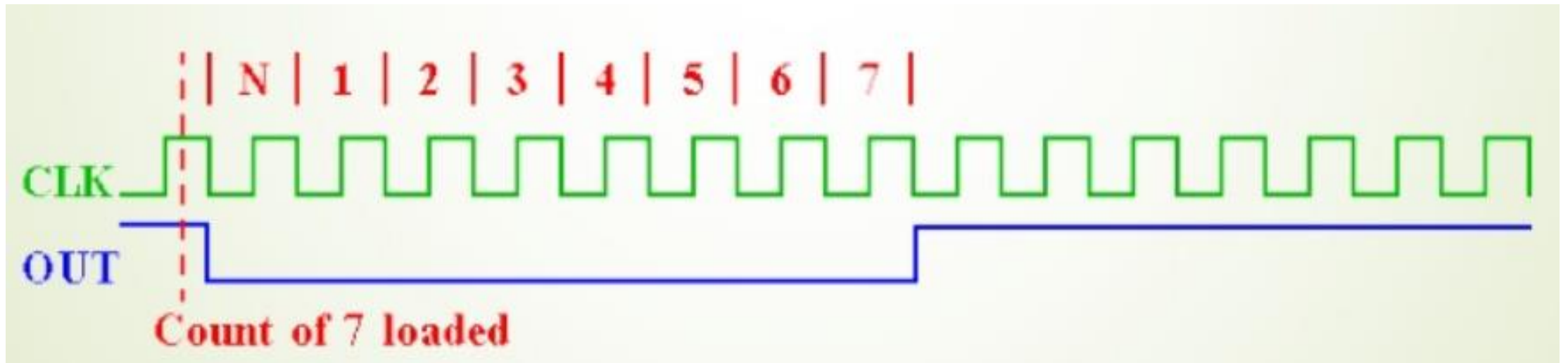
0	0	0	Mode 0
0	0	1	Mode 1
X	1	0	Mode 2
X	1	1	Mode 3
1	0	0	Mode 4
1	0	1	Mode 5

BCD

0	Binary Counter 16-bits
1	Binary Coded Decimal (BCD) Counter (4 Decades)

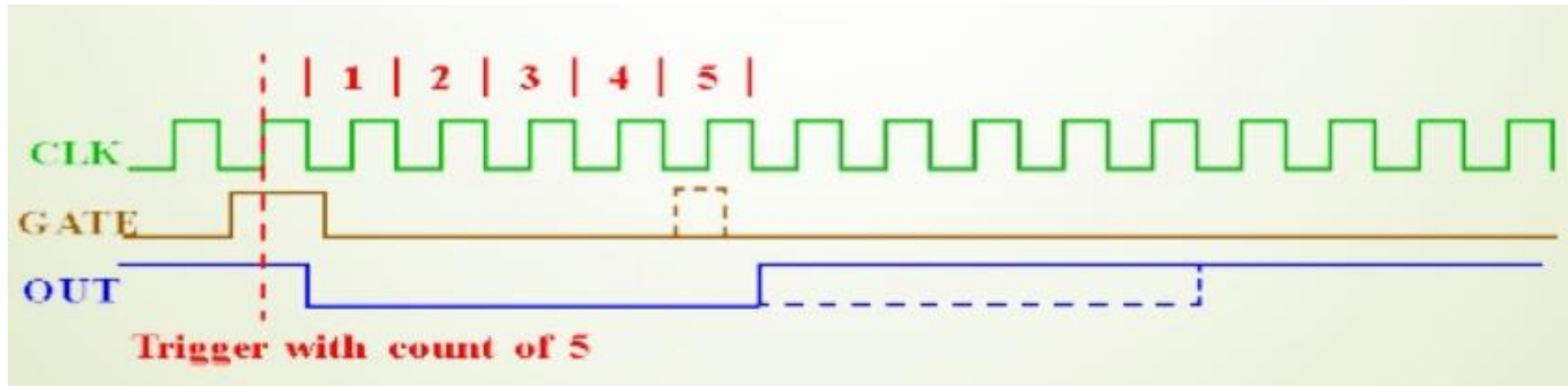
Mode 0

- Mode 0: Interrupt on terminal count - The output becomes a logic 0 when the control word is written - and remains there until N plus the number of programmed counts



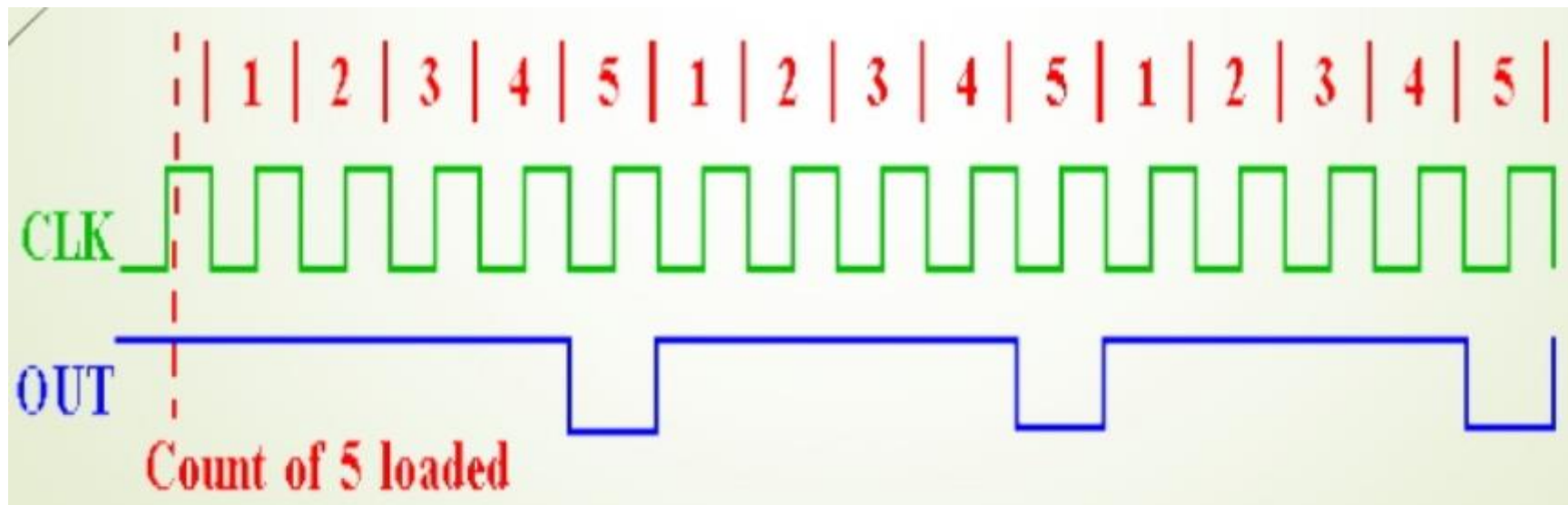
Mode 1

- MODE 1:- HW triggered / programmable one shot. The triggering must be done through the GATE input by sending a 0-to-1 pulse to it.
 - 1) Load the count register
 - 2) A 0-to-1 pulse must be sent to the GATE input to trigger the count. 3) In Mode 1, after sending the 0-to-1 pulse to GATE, OUT becomes low and stays low for a duration of $N \cdot T$, then becomes high and stays high until the GATE is triggered again.
 - 4) If during the activation, a retriggered happened, then restart the down counting.



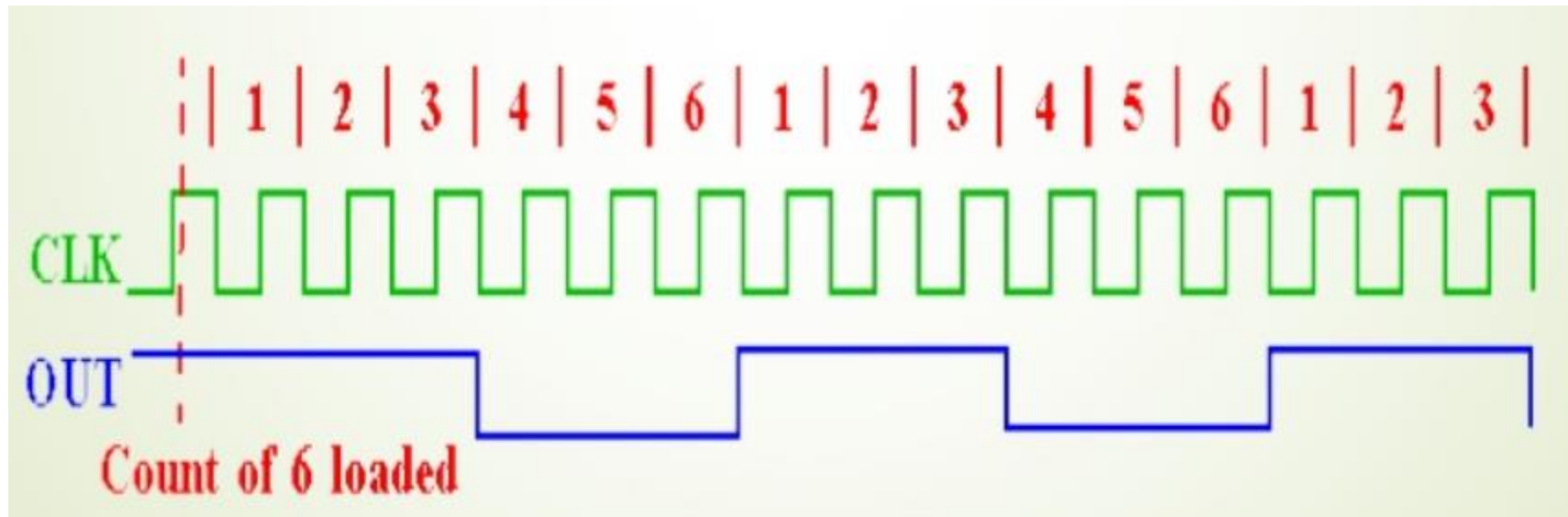
Mode 2

- Mode 2: Rate Generator (Divide-by-N counter) In Mode2,
- if GATE=1, OUT will be high for $N \cdot T$, goes low only for one clock pulse, then counter is reloaded automatically,
- The cycle is repeated until reprogrammed or G pin set to 0.



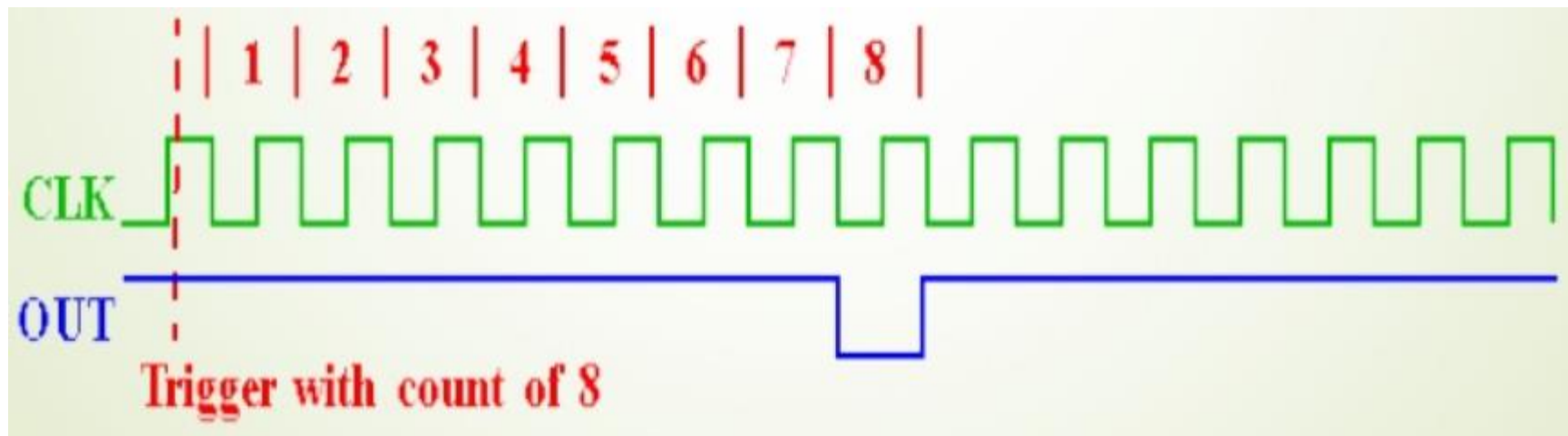
Mode 3

- Mode 3: Square wave rate generator
- Mode 3: Generates a continuous square-wave with G set to 1.
- if count is even, 50% duty cycle otherwise OUT is high 1 cycle longer



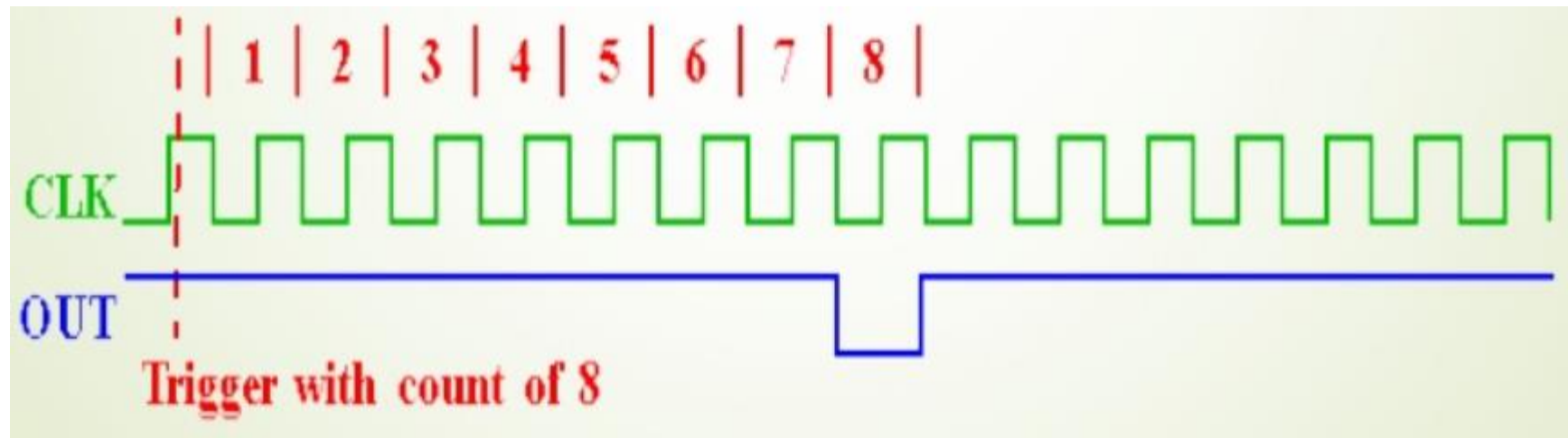
Mode 4

- Mode 4: Software triggered strobe
- In Mode4, if GATE=1, the output will go high when loading the count, it will stay high for duration $N \cdot T$.
- After the count reaches zero, it becomes low for one clock pulse, then goes high again and stays high until a new command word or new count is loaded
- To repeat the strobe, the count must be reloaded



Mode 5

- Mode 5: Hardware triggered strobe
- Similar to Mode4, except that the triggering must be done with the GATE input
- The count starts only when a 0-to-1 pulse is sent to the GATE input
- If GATE retriggered during the counting, it will restart the down counting



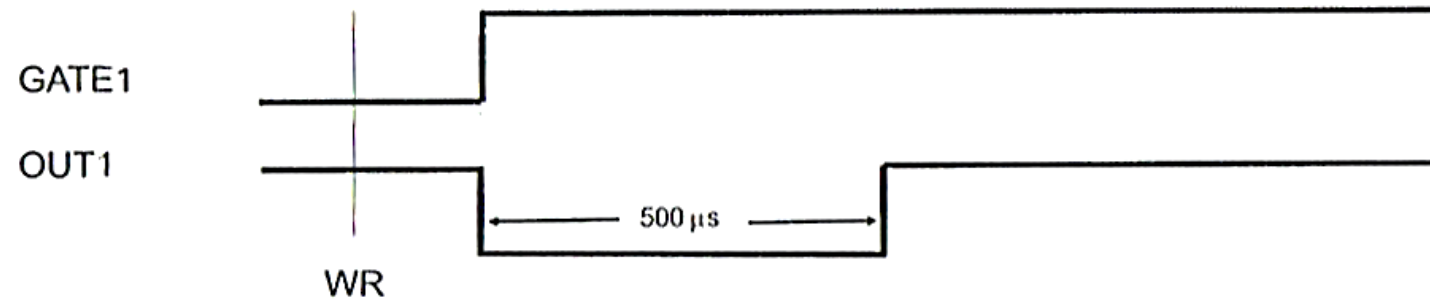
Example

Example 5-9

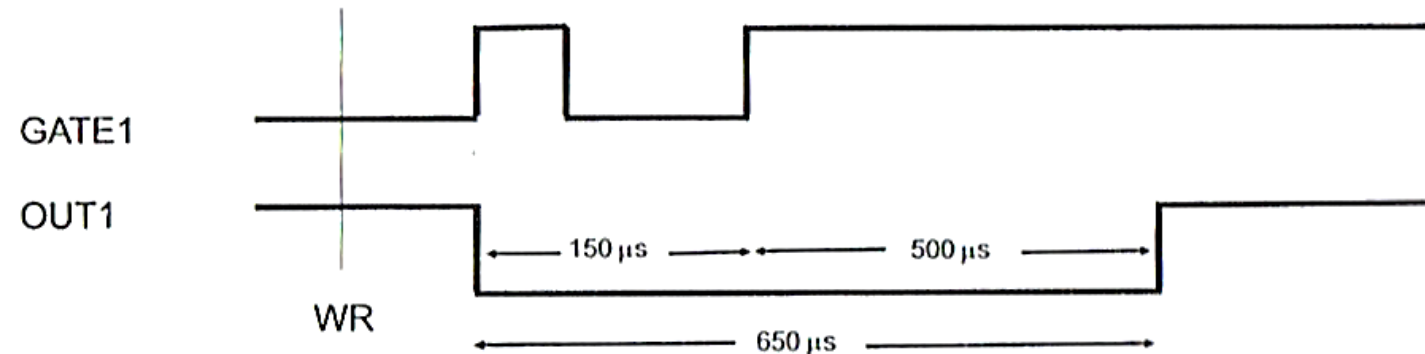
- (a) If $\text{CLK1} = 1 \text{ MHz}$ and $N = 500$, show the output for OUT1 if it is programmed for mode 1.
 (b) Assume that after 150 clock pulses, GATE1 is retriggered. Show the output for OUT1.

Solution:

- (a) Notice that OUT1 becomes low only when GATE1 goes from 0 to 1.



- (b) If GATE1 is retriggered after 150 clock pulses, COUNT1 is reloaded with $N = 500$ and the count starts all over again, making the OUT1 pulse duration $650 \mu\text{s}$ as shown next.



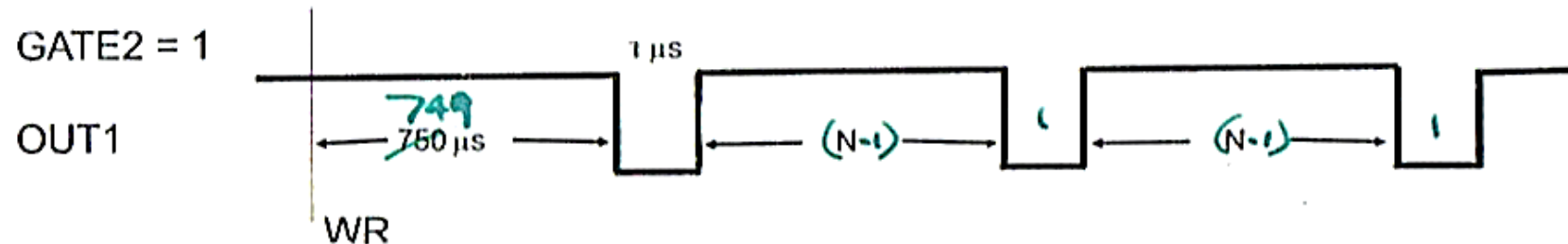
Example

Example 5-10

If $\text{CLK2} = 1 \text{ MHz}$, $\text{GATE2} = 1$, and $N = 750$, show OUT2 if COUNT2 is programmed for mode 2.

Solution:

Notice that the count is reloaded automatically and the counter continues to produce OUT2 .



Mode 3: square wave rate generator

Example

Example 5-11

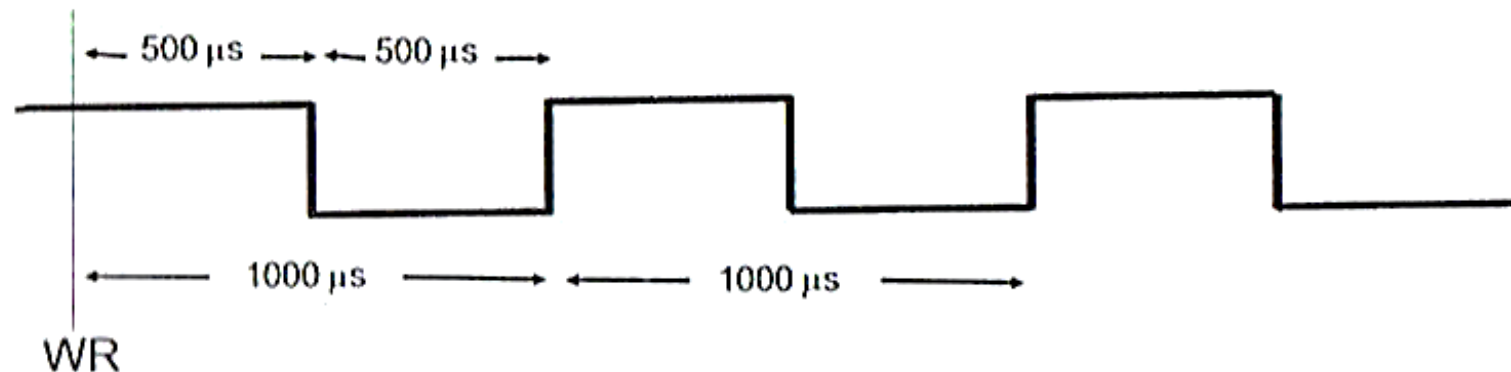
If $\text{CLK2} = 1 \text{ MHz}$, $\text{GATE1} = 1$, $N = 1000$, show OUT1 if COUNT1 is programmed for mode 3.

Solution:

Since the clock period is $1 \mu\text{s}$, OUT1 is high for $500 \mu\text{s}$ and low for $500 \mu\text{s}$, producing the square wave of 1 ms period continuously, as shown next.

$\text{GATE1} = 1$

OUT1



Mode 4: software triggered strobe

Example

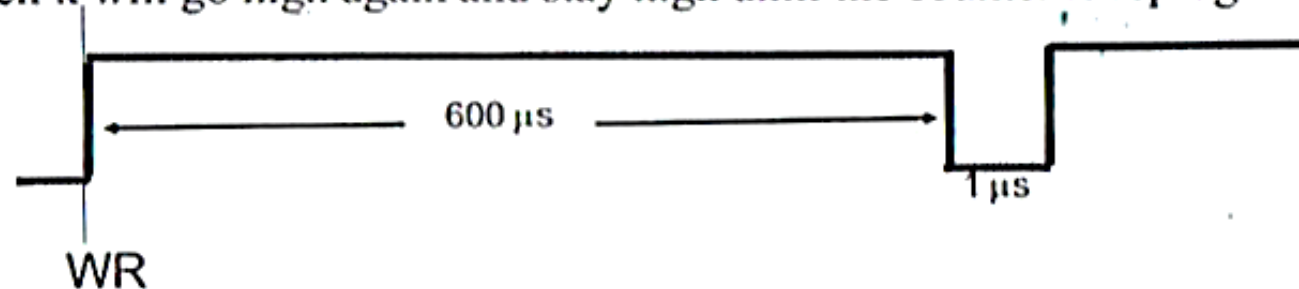
Example 5-12

If $\text{CLK0} = 1 \text{ MHz}$, $\text{GATE0} = 1$, and $N = 600$, show the shape of OUT0 where counter 0 is programmed for mode 4.

Solution:

Since the CLK0 period is $1 \mu\text{s}$, after the count is loaded OUT0 will be high for $600 \mu\text{s}$ and will go low for $1 \mu\text{s}$. Then it will go high again and stay high until the counter is reprogrammed, as shown below.

$\text{GATE0} = 1$
 OUT0



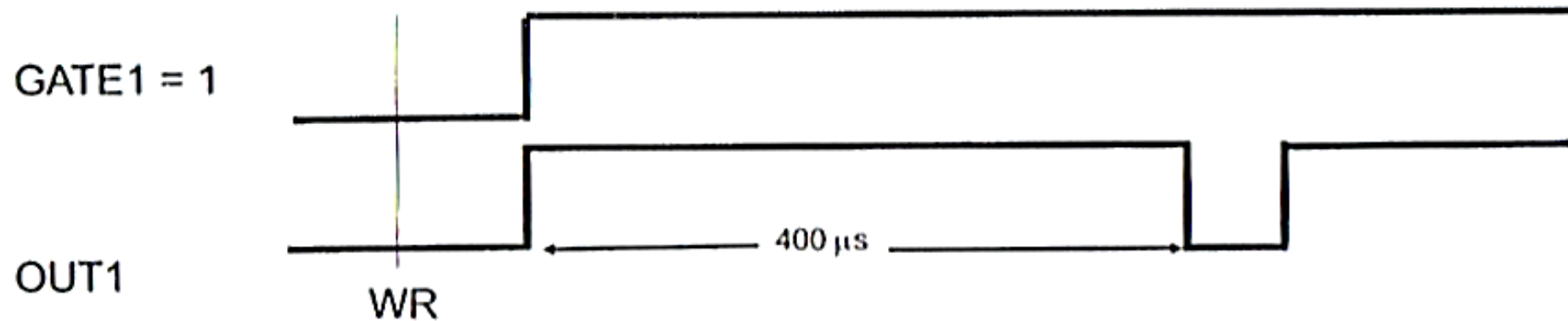
Example

Example 5-13

If $\text{CLK1} = 1 \text{ MHz}$, and $N = 400$, show the output for OUT1 if it is programmed for mode 5.

Solution:

Notice that the count starts only when the 0-to-1 pulse is applied to GATE1.



Mode 5: hardware triggered strobe

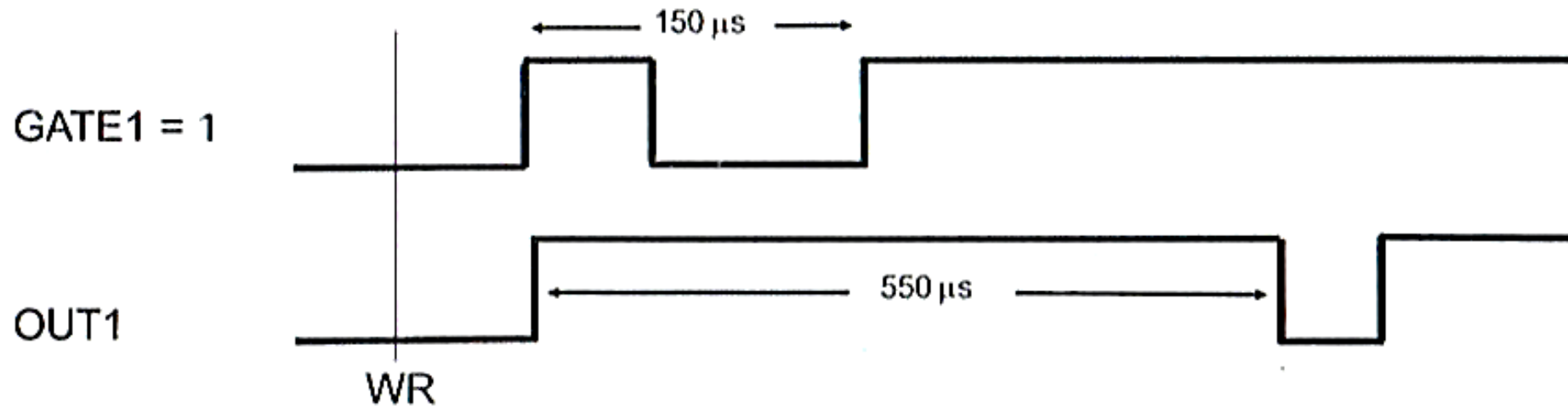
Example

Example 5-14

In Example 5-13, assume that GATE1 is retriggered after 150 pulses. Show the output for OUT1.

Solution:

If GATE1 is retriggered after 150 clock pulses into the countdown, COUNT1 is reloaded with $N = 400$ and the counts begins again, making the OUT1 pulse duration $550 \mu\text{s}$, as shown next.



Programming of 8254 Programmable Interval Timer

- Outline
 - Programming of 8254 Programmable Interval Timer

Example 2: Write a program to generate a square wave of 1KHz frequency on OUT 1 pin of 8253/54. Assume CLK1 frequency is 1MHz and address for control register = 0BH, counter 1 = 09H and counter 2 = 0AH.

Sol. : To get square wave mode 3 is selected count should be $\frac{1 \text{ MHz}}{1 \text{ KHz}} = 1000$

Control word :

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
SC ₁	SC ₂	RW ₁	RW ₀	M ₂	M ₁	M ₀	BCD
0	1	1	1	0	1	1	1

Count= Input Frequency/ Desired Frequency

= 77H

Source program

```

MOV  AL, 77H
OUT  0BH, AL           ; Loads control word (77H) in
                        ; the control register.

MOV  AL, 00H           ; loads lower byte (00) of the count
OUT  09H, AL

MOV  AL, 10            ; Loads higher byte (10) of the count
OUT  0AH, AL

```


Example Write instruction for 1KhZ square wave at Ctr 1,
Address of counter 0 =80H and input clock frequency is 0.5 MHZ

- Control word = 76H

D7	D6	D5	D4	D3	D2	D1	D0
SC1	SC2	RW1	RW0	M2	M1	M0	BCD
01		Load 16 bit (11)		011 (mode 3)			0

- Count= $0.5 \text{ MHz} / 1 \text{ KHZ} = 2000 = 07D0H$
- Instructions
 - MOV AL,76H ; load Control word for Ctr 1 mode 3
 - OUT 83H,AL ; write to Ctrl reg
 - MOV AL, D0H; lower order byte cnt
 - OUT 81H,AL
 - MOV AL,07H ; higher order byte
 - OUT 81H,AL
 - HLT

8254 Timer Example

- To program a given counter to divide the CLK input frequency - one must send the divisor to that specific counter's register.
- Although all three counters share the same control register, the divisor registers are separate for each counter
- Example: given the port addresses for 8253/54:
 - Counter 0: 94H
 - Counter 1: 95H
 - Counter 2: 96H
 - Control Reg: 97H

Example:

- **Task1:** program counter 0 for binary counter for mode 3 to divide CLK0 by number 4282 (BCD)

```
MOV AL, 0011 0111B
OUT 97H, AL
MOV AX, 4282H (BCD needs H)
OUT 94H, AL (Low Byte)
MOV AL, AH
OUT 94H, AL (High Byte)
OUT0 = CLK0 / 4282
```

Generating a Waveform with the 8254.

- Program a 8254 timer connected to function at I/O ports 0700H, 0702H, 0704H, and 0706H of an 8086 microprocessor to generates a 100 KHz square-wave at OUT0 and a 200 KHz continuous pulse at OUT1. The input clock frequency is 8 MHZ.
- Counter 0 uses mode 3
- Counter 1 uses mode 2.

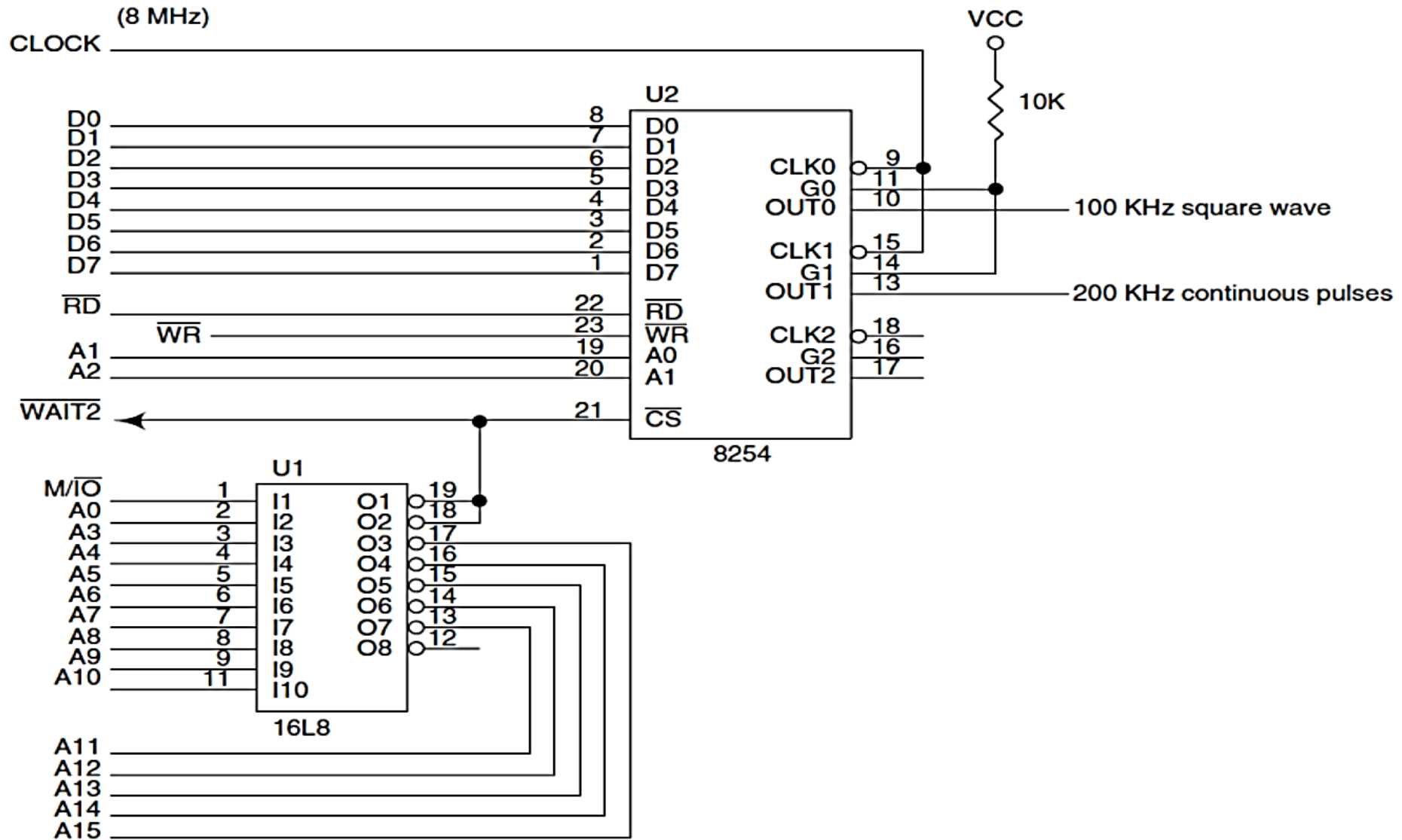


Figure 1: The 8254 interfaced to an 8 MHz 8086 so that it generates a 100 KHz square wave at OUT0 and a 200 KHz continuous pulse at OUT1.

EXAMPLE 11-24

;A procedure that programs the 8254 timer to function as illustrated in Figure 1

```
TIME PROC NEAR USES AX DX
MOV DX,706H ;program counter 0 for mode 3
MOV AL,00110110B
OUT DX,AL
MOV AL,01110100B ;program counter 1 for mode 2
OUT DX,AL
MOV DX,700H ;program counter 0 with 80
MOV AL,80
OUT DX,AL
MOV AL,0
OUT DX,AL
MOV DX,702H ;program counter 1 with 40
MOV AL,40
OUT DX,AL
MOV AL,0
OUT DX,AL
RET
TIME ENDP
```

Homework

- Write an assembly code to generate a continuous pulse of 50 KHZ